

Unit 14, Lesson 2: Interpreting Joint and Marginal Frequencies



Benchmarks

MA.912.DP.3.1 Construct a two-way frequency table summarizing bivariate categorical data. Interpret joint and marginal frequencies and determine possible associations in terms of a real-world context.

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate, and interpret the different components and quantities in the display.



Learning Targets

- ☐ I can determine joint and marginal frequencies.
- ☐ I can interpret joint and marginal frequencies in a real-world context.
- ☐ I can interpret a graphical display for bivariate categorical data.



14.2.1 Warm-Up: Bell Work

1. A survey conducted at Ambidextrous High School asked all 1,700 students to indicate their grade level and if they are left-handed or right-handed. Only 59 of the 491 freshmen are left-handed. Out of the 382 students in the sophomore class, 289 of them are right-handed. There are 433 students in the junior class and 120 of them are left-handed. There are 307 right-handed seniors.

Complete the two-way frequency table to display the results of the survey.

Dominant Hand Survey

					Total
Right-Handed					
Left-Handed					
Total					



14.2.2 Collaboration: Joint and Marginal Frequencies

1. In 2016, the Summer Olympics were held in Rio de Janeiro, Brazil. The United States of America won the most medals with a total of 121. China came in second, winning 70 total medals. This was a much higher total than the 9 medals China won two years later in the 2018 Winter Olympics held in Pyeongchang, South Korea. Norway led the medal count in 2018 with 39, a much higher number than the 4 total medals Norway won two years earlier in Brazil. Canada came in third in the 2018 Winter Olympics, winning 29 total medals.
 - a. Work with your partner to use the data regarding medal counts from the 2016 Summer Olympics and the 2018 Winter Olympics to make a two-way frequency table showing how the different countries' medal counts compare to each other.

	U.S.	China	Norway	Canada	Total
2016 Summer Olympics					217
2018 Winter Olympics					
Total	144				317

- b. How many total medals did the U.S. win in the 2016 and 2018 Olympics combined?

- c. Which team won the most medals in the two Olympics combined?
 - d. How many medals did Norway win in the 2018 Olympics?
 - e. Complete the statement.

_____ won the most medals in either the 2016 or 2018 Olympics, winning _____ medals in those games.

- f. What does the joint relative frequency, $\frac{121}{317}$, mean in this context?

2. Work with your partner to write questions that can be answered with the data in the table.
 - a. Write at least one question that must be answered with a joint frequency.
 - b. Write at least one question that must be answered with a marginal frequency.



14.2.3 Try It: Mono Lake

In California, there is a saltwater lake called Mono Lake. The salinity, or the concentration of salt in the lake, has varied over time. According to the Global Nature Fund, the salinity of Mono Lake was at an all-time high of 99 grams per liter (g/l) in 1982. The concentration of salt in Mono Lake increased as a result of water diversion. Streams that feed Mono Lake were diverted, and the fresh water was used for drinking water.

Regulations and compromises have helped control water diversion at Mono Lake since then, and the salinity of the lake is slowly returning to normal. In 2008, there were 79.8 grams of salt per liter in Mono Lake. As a result, the populations of two main organisms that live in the lake, brine shrimp and alkali flies, have increased to more optimal levels. In 1982, there were an estimated 2,000 brine shrimp per liter of lake water. In 2008, there were 3,000 brine shrimp per liter of lake water. In 1982, there were 1,200 alkali flies per liter of lake water. That number nearly tripled to 3,500 alkali flies per liter in 2008 when the salinity of Mono Lake was 79.8 g/l.

1. Construct a two-way frequency table that shows the alkali flies per liter of lake water and brine shrimp per liter of lake water living at Mono Lake when the salinity was 99 g/l and 79.8 g/l.

	Salinity of 99 g/l	Salinity of 79.8 g/l	Total
Brine Shrimp			
Alkali Flies			
Total			

2. Work with your partner to answer each question.

- a. How many alkali flies per liter of lake water were found in Mono Lake when the salinity was 99 g/l?
- b. When the salinity of Mono Lake was 79.8 g/l, how many brine shrimp and alkali flies per liter of lake water were found?
- c. Which is greater — the combined total of alkali flies, or the combined total of brine shrimp in Mono Lake from these two samples?
- d. What was the salinity of Mono Lake when the population of alkali flies was 3,500 per liter?
- e. What was the total amount of brine shrimp found in Mono Lake from the two studies?
- f. Was the population of brine shrimp smaller when the salinity of Mono Lake was 99 g/l or 79.8 g/l?



14.2.4 Your Turn!

1. At Dupont Middle School in Jacksonville, FL, 635 students were asked which social media site they updated the most on a daily basis. Seventy-five seventh graders said they used Instagram the most. A total of 321 students said they updated Facebook the most. Of the 247 eighth graders, 23 of them updated Instagram the most. There are 181 sixth graders at Dupont. Of the 158 Twitter users, 45 of them were sixth graders. There were 123 eighth graders using Facebook.

Complete the two-way table summarizing the data.

Dupont Middle School Social Media Preference

	Facebook	Instagram	Twitter	Total
6 th				
7 th				
8 th				
Total				

2. Austin studies birth weight and height as part of his research at Florida School of Traditional Midwifery in Gainesville, Florida. The following two-way table shows data from births in the month of June 2021 at UF Health Shands Hospital.

Births from UF Health Shands Hospital (June 2021)

	Less Than 5 lbs.	5-8 lbs.	More Than 8 lbs.	Total
Less Than 16 in.	21		2	32
16-20 in.	14	97	68	
More Than 20 in.	0	8		
Total			101	

Use the table to answer the following questions and prompts.

- Circle the smallest marginal frequency in the table.
- Underline the largest joint frequency in the table.
- Describe what the value 14 means in context of the data.
- Describe what the value 101 means in context of the data.



14.2.5 Wrap-Up: Ticket Out the Door

Use the two-way table showing students' field trip preferences to answer the questions.

Students' Field Trip Preferences

	Amusement Park	Water Park	Mystery Destination	Total
Freshmen	25	9	20	54
Sophomores	35	6	5	46
Total	60	15	25	100

- Underline the smallest joint frequency.
- Explain what the smallest joint frequency represents in terms of students' field trip preferences.
- Circle the largest marginal frequency.
- Explain what the largest marginal frequency represents in terms of students' field trip preferences.